5.1.2 Population genetics and epigenetics

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the role of natural selection in changing allele frequencies within populations	To include the link between malaria and the frequency of the sickle cell allele including the effect on the phenotype of each of the three possible genotypes for the normal and sickle cell allele. *M1.4, M1.5, M2.2, M2.3, M2.4*
(b)	the link between the changes in the amino acid sequence to the change in structure and properties of proteins (e.g. haemoglobin)	HSW1, HSW5, HSW6
(c)	the use of Hardy-Weinberg equations to analyse changes in allele frequencies in populations	The equations for the Hardy-Weinberg principle will be provided where needed in assessments and do not need to be recalled $p^2+2pq+q^2=1$ $p+q=1$ HSW3, HSW8 *M2.1, M2.2, M2.3, M2.4*
(d)	factors other than natural selection that contribute to genetic biodiversity	To include the role of the founder effect and genetic bottlenecks in creating genetic differences between human populations. Examples to include blood group distribution and Ellis-van Creveld syndrome distribution. HSW1, HSW5, HSW6, HSW9, HSW10, HSW12

(e)	the role of geographical and reproductive isolation in the formation of new species	To include a consideration of the implications for speciation of primates, including humans.
(f)	epigenetics in terms of the effect of environment on gene expression.	To include theories of the role of DNA methylation and histones in gene expression **AND** a review of some human epigenetic studies (such as the Norrbotten studies, studies on the effect of the Dutch Hunger Winter and twin studies) and possible implications from these studies. *M1.7* HSW1, HSW2, HSW5, HSW6, HSW8, HSW9, HSW10, HSW11, HSW12